



INSTITUTE OF ENGINEERING & MANAGEMENT
SALT LAKE, KOLKATA
Affiliated to MAKAUT and approved by AICTE



DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING
POWER SYSTEM LABORATORY-II

CODE: PCC-EEE 691

ASSIGNMENTS

Title: Performance analysis and testing of polarity, ratio and magnetization characteristics of CT and PT for proper transformer connection in a system. Design and simulation of connection diagram by CT-PT for protection of a distribution system using MATLAB/ETAP.

Assignments. 1. Performance analysis and testing of polarity, ratio and magnetization characteristics of CT and PT for proper transformer connection in a system using MATLAB and Model and analyze the saturation characteristics of CTs and PTs under different loading conditions using MATLAB Simulink.

2. Analyze the behavior of CTs and PTs when subjected to distorted waveforms using MATLAB. Investigate the effect of harmonics on their accuracy.

Title: Testing of time-dependent voltage-current characteristics of over current relay for application to generator protection, transformer protection and motor protection. Design of op-amp based overcurrent fault protection.

Assignments. 1. Model an overcurrent relay with CT inputs and an undervoltage relay with PT inputs in MATLAB Simulink.

2. Test various fault scenarios to analyze the relay operation.

Title: Testing of IDMT characteristics of under voltage relay and MATLAB-based design for application in transmission line faults analysis.

Assignments. 1. Test the IDMT Characteristics of an Undervoltage Relay

- Implement an Inverse Definite Minimum Time (IDMT) relay model in MATLAB/Simulink.
- Simulate voltage sags to test relay operation and trip time.

Title: Design and simulation of transient stability of a multi-machine system in a power network for reliable power supply using MATLAB.

Assignments. 1. A swing curve shows the variation of the rotor angle of a synchronous generator with respect to time after a disturbance. Write a code for simulating a swing curve for a power system after a fault is cleared and analyze the output. (input data can be considered as per standard and compatible values)

2. A swing curve is computed based on the defined parameters, considering the varying dynamic loads and the rotor angle response is calculated by accounting for different stages of fault clearance and sinusoidal variations. This allows for a comprehensive understanding of how dynamic loads impact the swing curve. Write a code for such dynamic load impact on power system swing curve by which engineers may realize system operation, load management, and control strategies. (input data can be considered as per standard and compatible values)

Title: Simulation and analysis of different types of unsymmetrical and symmetrical faults exists in distribution system network using MATLAB.

Assignments. 1. Harmonic Analysis of a Distribution System using MATLAB

2. Fault analysis using overcurrent relay protection in matlab simulink in power system

Title: Simulation of different IEEE distribution bus systems framework and assessment of voltage, current and load profile using AC load flow analysis implementing Gauss-Seidel method in ETAP/MATLAB platform.

Assignments. 1. Simulation of different IEEE distribution bus systems framework and assessment of voltage, current and load profile using AC load flow analysis implementing Newton Raphson method in ETAP/MATLAB platform.

2. Development of bus admittance matrix of an interconnected power network for load flow analysis following IEEE standard using MATLAB. Design and AI based simulation of a microgrid system using renewable and non-renewable micro-sources for demand side management (DSM).

Title: Design of protection scheme of distribution substation using earth fault relay and testing of its IDMT characteristics.

Assignments. 1. Testing of time-dependent voltage current characteristics of Static relay and simulation of protection scheme using Static relay arrangement given to a distribution network either for bus-bar or feeder protection.

2. Testing of time-dependent characteristics of on-time delay relay and off-time delay relay for application of load management. Design of simulation model for activation or deactivation of heavy electrical loads to reduce peak demand using on-time delay relay and off-time delay relay.